

**Attachment 1**



**AURORA ENERGY PTY LTD**

**ABN 85 082 464 622**

**Network Tariff Application Guide**

**Period 1 2008**  
**(Version 1.1)**

**To apply on and from 1 January 2008**  
**(or as subsequently updated)**

### Record of Amendments

| Issue No | Date     | Comments                                                                          | Author                        |
|----------|----------|-----------------------------------------------------------------------------------|-------------------------------|
| 01       | Oct. '07 | NW#267308-v1                                                                      | Eric Lown/<br>Peter Milbourne |
| 02       | Dec. '07 | NW#271141-v1 Period 1 2008<br>Network Tariff applications for 2008 calendar year. | Eric Lown/<br>Peter Milbourne |
|          |          |                                                                                   |                               |
|          |          |                                                                                   |                               |

### Disclaimer

This document is for generating internal and external information belonging to Aurora Energy acting in its capacity as a Distribution Network Service Provider in the state of Tasmania only and it may or may not contain all available information on the subject matter this document purports to address. The information contained in this document is subject to review and Aurora Energy may amend this document at any time. Amendments will be indicated in the Amendment Table, but Aurora Energy does not undertake to keep this document up to date.

To the maximum extent permitted by law, Aurora Energy makes no representation or warranty (express or implied) as to the accuracy, reliability, or completeness of the information contained in this document, or its suitability for any intended purpose. Aurora Energy (which, for the purposes of this disclaimer, includes all of its related bodies corporate, its officers, employees, contractors, agents and consultants, and those of its bodies corporate) shall have no liability for any loss or damage (be it direct or indirect, including liability by reason of negligence or negligent misstatement) for any statements, opinions, information or matter (expressed or implied) arising out of, contained in, or derived from, or for any omissions from, the information in this document.

### CONTACT

This document is the responsibility of the Regulatory Group within the Network Division of Aurora Energy Pty Ltd. (ABN 85 082 464 622)

Please contact the indicated owner of the document with any inquiries.

Mr Leigh Mayne  
Aurora Energy Pty Ltd  
Level 1, 177 Main Road  
Moonah Tasmania 7008  
Ph: (03) 6271 6544

In this Application Guide words and phrases printed like **this** are defined terms and listed under Definitions in this document.

### REVIEW DATE

Annually.

## Table of Contents

### Cover Page

|                                                                                        |    |
|----------------------------------------------------------------------------------------|----|
| <b>1.0 Application of Network Tariffs</b> .....                                        | 4  |
| (a) Goods and Services Tax (GST)                                                       |    |
| (b) Choice of Network Charges                                                          |    |
| (c) Mid Month Change of “Retailer of Choice”                                           |    |
| (d) Stand By Electricity Supply                                                        |    |
| <b>2.0 Published Network Tariffs</b> .....                                             | 5  |
| (a) General Network [N01].....                                                         | 6  |
| (b) General Network - Business [N02].....                                              | 6  |
| (c) Low Voltage Metered Demand (kW) [N03].....                                         | 6  |
| (d) Uncontrolled Energy [N05].....                                                     | 6  |
| (e) Controlled Energy [N06].....                                                       | 6  |
| (f) Small Low Voltage Unmetered [N07].....                                             | 7  |
| (g) Low Voltage Day/Night (Irrigation) [N08].....                                      | 7  |
| (h) Low Voltage Metered Demand (kVA) [N09].....                                        | 7  |
| (i) Low Voltage Time of Use (Interval) [N12].....                                      | 7  |
| (j) Low Voltage Pay as You Go [N13].....                                               | 8  |
| (k) Low Voltage Specified Demand [N16].....                                            | 8  |
| (l) High Voltage Metered Demand (kVA) [N10].....                                       | 8  |
| (m) High Voltage Metered Demand (kW) [N11].....                                        | 8  |
| (n) High Voltage Specified Demand [N17].....                                           | 9  |
| (o) High Voltage Installations with Specified Demand<br>in excess of 2.0MVA [N15]..... | 9  |
| (r) Street Lighting [N20].....                                                         | 9  |
| <b>3.0 Individually Calculated Customer Tariffs</b> .....                              | 9  |
| <b>4.0 Reassignment of Network Tariffs</b> .....                                       | 10 |
| <b>5.0 Maximum Demand Application</b> .....                                            | 11 |
| (a) Changes to Maximum Demand                                                          |    |
| (b) Increases in Maximum Demand                                                        |    |
| (c) Temporary Increases in Maximum Demand                                              |    |
| <b>6.0 Specified Demand Application</b> .....                                          | 12 |
| <b>7.0 Definitions</b> .....                                                           | 12 |
| <b>8.0 References</b> .....                                                            | 13 |

### 1.0 Application of Network Tariffs

#### Goods and Service Tax (GST)

The network use of system prices and regulated network tariffs listed in the schedules, unless otherwise stated, are **exclusive** of GST.

#### Choice of Network charges

It is a matter for the customer's *Retailer of Choice* to determine the network tariff that is appropriate for their needs. Transfers between network tariffs will apply from the beginning of the normal billing period following the month in which written advice is received and transfers will not have any retrospective effect.

*Retailers of Choice* and customers who choose to transfer from an obsolete tariff will lose all rights to all Obsolete Tariffs at that premise, i.e. the entire site will be required to move to currently available published network tariffs. Aurora Energy reserves the right to review a customer's network tariff in the event of any electrical load changes and will notify the customer's *Retailer of Choice* regarding any resulting network tariff pricing changes. The final decision on the appropriate network tariff shall be at the discretion of Aurora Energy.

There will be instances where a customer may have a separate connection agreement with the network service provider (Aurora Energy) and in these instances Aurora may directly invoice the customer for network use and the customer's *Retailer of Choice* will only provide energy related commercial services.

#### Mid Month Change of Retailer of Choice

When a customer elects to change their *Retailer of Choice* at any time during a calendar month, *Network Use of System (NUoS)* invoices will be allocated between the retailers involved up to and from the actual date the transfer was completed in the market.

*NUoS* invoices will use the "actual readings" obtained on the day of transfer.

Unmetered electricity supplies will be transferred on a pro rata basis based on the number of days in the month the customer was contracted with each retailer. It will not be possible to transfer unmetered electricity supplies between Retailers on the same business day.

#### Standby Electricity Supply

Where customers with critical electricity supply needs have a requirement for standby electricity supply capability, applicable network charges will be negotiated on the basis of the assets involved and the amount of feeder capacity required to be kept in reserve to accommodate the standby supply.

#### Embedded Generation

Network Use of System (NUoS) charges for *Embedded Generation* will be individually calculated. (Refer to Section 3.0)

### 2.0 Published Network Tariffs

Published Network Tariff applications depend on a number of facets including annual electrical energy consumption measured at the *connection point*, the supply voltage measured at the *connection point*, the method of connection to Aurora's *distribution system* and the type of meter(s) installed at the customer's installation.

Aurora Energy's current published Network Tariff prices can be found on our website at:

[http://www.auroraenergy.com.au/powerline\\_network/quicklinks/network\\_tariffs.asp](http://www.auroraenergy.com.au/powerline_network/quicklinks/network_tariffs.asp).

**Aurora's Network Tariff applications are:**

| Description                                            | Aurora Code | MSATS Code  | Type                      |
|--------------------------------------------------------|-------------|-------------|---------------------------|
| General Network - Residential                          | N01         | AuResGen    | Published Tariff          |
| General Network - Business                             | N02         | AuBLVGen    | Published Tariff          |
| Low Voltage Metered Demand (kW)                        | N03         | AuBLVDmKW   | Published Obsolete Tariff |
| Uncontrolled Heating Tariff                            | N05         | AuHeatUnco  | Published Tariff          |
| Controlled Heating Tariff                              | N06         | AuHeatCont  | Published Tariff          |
| Small Low Voltage Unmetered                            | N07         | AuUMS       | Published Tariff          |
| Low Voltage Day/Night (Irrigation)                     | N08         | AuIrrig     | Published Tariff          |
| Low Voltage Metered Demand (kVA)                       | N09         | AuBLVDMKVA  | Published Tariff          |
| Low Voltage Time of Use (Interval)                     | N12         | AuResLVTuEn | Published Tariff          |
| Low Voltage Pay as You Go                              | N13         | AuPAYG      | Published Tariff          |
| Low Voltage with Specified Demand                      | N16         | AuCLVDm2    | Published Tariff          |
| High Voltage Metered Demand (kVA)                      | N10         | AuBHVDmKVA  | Published Tariff          |
| High Voltage Metered Demand (kW)                       | N11         | AuBHVDmKW   | Published Obsolete Tariff |
| High Voltage with Specified Demand                     | N17         |             | Published Tariff          |
| High Voltage with Specified Demand in excess of 2.0MVA | N15         | AuCHVDm2    | Published Tariff          |

## Aurora Energy Network Tariff Application Guide

|                                                    |     |  |                       |
|----------------------------------------------------|-----|--|-----------------------|
| Street Lighting                                    | N20 |  | Published<br>Tariff   |
| <b>Individually Calculated Customers<br/>(ICC)</b> |     |  | Negotiated<br>Tariff. |

### **General Network [N01]**

Available for electricity consumption in premises used wholly or principally as private residential dwellings where an energy only meter is installed, known as a Type 5 or 6 metering installation, as per the *National Electricity Rules*.

A private residential dwelling is a house, flat, home unit, town house or similar qualifying residential premise.

**(MSATS Code – AuResGen)**

### **General Network - Business - [N02]**

Available for low voltage supplies to premises used for any non-domestic purpose where an energy only meter is installed, known as a Type 6 metering installation, as per the *National Electricity Rules*.

**(MSATS Code – AuBLVGen)**

### **Low Voltage Metered Demand (kW) [N03]**

Applies to a customer taking electricity supply at less than 1.0 kV.

A NEM compliant Type 1-4 interval meter is required for these installations.

This tariff is obsolete with no new connections allowed from 1 January 2005.

Customers who choose to transfer off an Obsolete Tariff will lose all rights to all obsolete tariffs on that premise, i.e. the entire site will be required to move to currently available published tariffs.

**(MSATS Code – AuBLVDmKW)**

### **Uncontrolled Energy Tariff [N05]**

Available for electricity consumption in commercial premises and also for premises used wholly or principally as private residential dwellings and may only be applied in association with tariff N01. The commercial application is for water heating only and the residential application can be for water heating and/or space heating. An energy only meter is installed, known as a Type 6 metering installation, as per the *National Electricity Rules*.

**(MSATS Code – AuHeatUnco)**

### **Controlled Energy Tariff [N06]**

Applicable to electricity that is separately metered and controlled by Aurora Energy for fixed wire appliances, heating systems, existing general purpose outlets that were previously connected to a controlled heating tariff, and may only be applied in association with tariffs N01 and N02. The minimum metering requirements are a Type 6 meter, switched by a timing device. The supply is to be available daily for a period of at least nine (9) hours between 2000 hours and 0700 hours the following day. A further period of two (2) hours is available daily between 1300 hours and 1630 hours. The selected times within these periods are to be at Aurora's discretion.

**(MSATS Code – AuEnCont)**

### **Small Low Voltage Unmetered Supplies [N07]**

From time to time there will be situations where an un-metered supply is required, the following types of connections can be charged as un-metered supply:

- Illuminated Street Signs.
- Public telephone kiosks.
- Electric fences.
- Two way radio transmitters.
- Fixed steady wattage installations
- Shows & public functions.
- Traffic lights.
- Level crossings.

Where a new un-metered supply is made available, the charge shall be based on the steady state wattage of the device/installation.

It is a requirement that an un-metered supply must have all components permanently connected.

**(MSATS Code – AuUMS)**

### **Irrigation [N08]**

A time of use application where all installed meters will be dual register types with the ability to record time of use and consumption. Energy is charged according to peak and off-peak time periods. The off-peak rate is available for a period of at least 10 hours between 8pm and 7am (EST) and all other times are peak times. A NEM compliant Type 5 meter installation is required for these installations.

**(MSATS Code – AuIrrig)**

### **Low Voltage Metered Demand (kVA) [N09]**

Applies to a customer taking electricity supply at less than 1.0 kV.

A NEM compliant Type 6 meter is required for these installations with the ability to measure both active and reactive power.

**(MSATS Code – AuBLVDMKVA)**

### **Low Voltage Time of Use Energy [N12]**

This tariff will be available from 1 July 2008 for electricity consumption in premises used wholly or principally as private residential dwellings with a *time of use* meter from which interval metering data can be obtained.

The time of use tariff provides customers with incentives to manage their electricity charges by moving their energy consumption from peak to shoulder or off-peak periods.

Aurora's "Controlled Energy Tariff [N06]" can be utilised in conjunction with "Low Voltage Residential Time of Use Energy" tariff [N12].

**(MSATS Code – AuResLVTuEn)**

### **Pay as You Go [N13]**

Available for electricity consumption in premises used wholly or principally as private residential dwellings where an energy only meter is installed, known as a Type 4 metering installation, as per the *National Electricity Rules*. A private residential dwelling is a house, flat, home unit, town house or similar qualifying residential premise.

This tariff will become obsolete from 30 June 2008 and will only be available to existing Aurora Pay As You Go customers from that date.

**(MSATS Code – AuPAYG)**

### **Low Voltage Specified Demand [N16]**

This tariff will be available from 1 July 2008 for large customers who are supplied directly from on-site substation terminals where there are no Aurora Energy distribution assets beyond the substation.

This is a published network tariff for customers who become contestable under the Electricity Supply Industry (Contestable Customer) Regulations 2005.

A customer can choose to move to the published tariff and elect to have a specified electrical demand. The tariff allows for a demand charge with a penalty for exceeding the specified demand for any billing period.

Each year Aurora will publish information relating to the application of Transmission Use of System (TUoS) related charges to the particular transmission node from which the customer receives electricity supply.

The individual distribution and metering charges are then added to the TUoS charge to achieve the total Network Use of System (NUoS) charge at the customer's connection point.

**(MSATS Code – AuCLVDm2)**

### **High Voltage Metered Demand (kVA) [N10]**

Applies to a Network User taking electricity supply at greater than 1.0 kV.

A NEM compliant Type 1-4 interval meter or a NEM compliant Type 6 meter is required for these installations with the ability to measure both active and reactive power.

Note: The [N10] tariff may be offered in the future as a 'Time of Use' application.

**(MSATS Code – AuBHVDmKVA)**

### **High Voltage Metered Demand (kW) [N11]**

Applies to a customer taking electricity supply at greater than 1.0 kV.

A NEM compliant Type 1-4 interval meter or a NEM compliant Type 6 meter is required for these installations.

This tariff is obsolete with no new connections allowed from 1 January 2005.

Customers who choose to transfer off an **obsolete** tariff will lose all rights to all obsolete tariffs on that premise, i.e. the entire site will be required to move to currently available published tariffs.

**(MSATS Code – AuBHVDmKW)**

### **High Voltage Specified Demand [N17]**

This tariff will be available from 1 July 2008 for customers who become contestable under the Electricity Supply Industry (Contestable Customer) Regulations 2005.

A customer can choose to move to the published tariff and elect to have a specified electrical demand. The tariff allows for a demand charge with a penalty for exceeding the specified demand for any billing period.

Each year Aurora will publish information relating to the application of Transmission Use of System (TUoS) related charges to the particular transmission node from which the contestable customer receives electricity supply.

The individual distribution and metering charges are then added to the TUoS charge to achieve the total Network Use of System (NUoS) charge for the customer's connection point.

### **High Voltage with a Specified Demand in excess of 2.0MVA [N15]**

This is a published network tariff for customers who become contestable under the Electricity Supply Industry (Contestable Customer) Regulations 2005.

Once contestable there is a grace period during which time customers can choose to remain with their existing supply arrangements for a defined period or they can choose to move to an available published tariff with a specified electrical demand. The tariff allows for a demand charge with a penalty for exceeding the specified demand for any billing period.

Each year Aurora will publish information relating to the application of Transmission Use of System (TUoS) related charges to the particular transmission node from which the contestable customer receives electricity supply. The published charges will be in the form of dollars per kVA of demand (\$'s/kVA) and there is no energy component.

The individual distribution and metering charges are then added to the TUoS charge to achieve the total Network Use of System (NUoS) charge at the customer's connection point.

**(MSATS Code – AuCHVDm2)**

### **Street Lighting [N20]**

The street lighting tariff rate is based on a "use of system charge" and is charged on a per lamp wattage rate. The rate does not include energy or light fitting charges. Light fitting charges are listed as a separate schedule.

## **3.0 Individually Calculated Customer (ICC) Network Prices**

Individually Calculated Customer (ICC) network prices will typically apply to customers with an electrical demand in excess of 2.0 MVA or where a customer's circumstances in a pricing zone identify the average shared network charge to be meaningless or distorted. Individually calculated customer network charges are determined by modelling the connection point requirements as requested by the customer or their agents. There also a need

for referral to appropriate Australian Standards when determining ICC network prices.

ICC prices are based on actual Transmission Use of System (TUOS) charges for the relevant transmission connection point, plus charges associated with the actual shared distribution network utilised for the electricity supply, plus connection charges based on the actual connection assets utilised. This provides the greatest cost reflectivity for this type of customer and is feasible since the number of such customers is relatively small.

ICC pricing is also justified by virtue of the shared distribution network assets being designated specifically to meet the requirements of these customers. Where the designation of shared network assets is difficult due to the specific connectivity of the customer, Aurora will apply ICC pricing on a mutually agreed basis.

ICC pricing can also be influenced by the **load factor** of the Customer's installation.

ICC pricing for customers with electrical demand that is less than 2.0 MVA could occur where:

- A customer has a dedicated supply system that is different and separate from the remainder of the supply network;
- There are only a small number of customers in a supply system making average prices inappropriate; or
- Inequitable treatment of otherwise comparable customers arises from the electrical demand lower limit of 2.0 MVA. Selection of these customers will be at Aurora's discretion.

Distribution network tariff charges for *Embedded Generation* will always be treated on an individually calculated basis.

### 4.0 Reassignment of Network Tariffs

Customers seeking tariff reassignment must remain on the reassigned tariff for a minimum twelve (12) month period. Exceptions will only be made at Aurora's discretion where it can be demonstrated that to not do so would result in unreasonable penalties or impose hardship on the customer.

This condition prevents customers from changing tariffs to take advantage of seasonal variations in prices according to their individual load; thereby bypassing payment that reflects use of the distribution network over a full twelve (12) month cycle.

### 5.0 Maximum Demand Application

For the purposes of determining a customer's maximum demand for Network Tariff application the following rules will apply:

#### **Definition of Maximum Demand**

*Maximum demand* means the electrical demand measured in kilo Volt Amps (kVA) or kilo Watts (kW) (now obsolete) and is calculated as the energy consumption recorded over the demand integration period divided by the demand integration period in hours. The demand integration period is thirty (30) minutes. The value of is used for setting the *maximum demand* charges to be paid by the customer to the distributor each billing period.

#### **Changes to Maximum Demand**

##### ***Increases in Electrical Demand***

Where a customer requires a change in Network Tariff due to an increase in electrical demand at a connection point, the customer must submit a written request to Aurora Energy. The increased level of electrical demand shall apply from the requested date or as near as possible thereafter, subject to any required works being completed by Aurora Energy.

##### ***Temporary Increases in Maximum Demand***

Temporary increases in electrical demand will:

- Be subject to negotiation and approval by Aurora Energy.
- Be defined in terms of "additional demand" for a specific period and charged at an agreed Demand Charge rate.
- Apply for one full billing period, except in the case of commissioning of new plant, in which case the duration of the temporary increase may be extended for the duration of the commissioning period.
- Be limited to one occurrence each twelve months or as otherwise agreed with Aurora Energy.

##### ***Reduction in Maximum Demand***

Where a customer requires a change in Network Tariff due to a reduction in electrical demand at their connection point, the customer must provide six (6) months written advice to Aurora Energy detailing their new requirement. Aurora Energy must notify the customer in writing within sixty (60) days advising the revised NUoS charges and also any costs associated with the relocation of Aurora's assets at their site.

However, following the installation by a customer of load management equipment approved by Aurora Energy or the implementation of a demand management initiative approved by Aurora Energy, the six (6) month notice period referred to above may be reduced at the discretion of Aurora Energy.

### 6.0 Specified Demand Application

Customers selecting published tariffs [N15], [N16] or [N17] are able to agree to a “Specified Demand” for their electrical installation. Once set this value of demand is charged for the next twelve months, if however the specified demand is exceeded the “excess charge” is applied to the demand in excess of the specified demand.

Renegotiation of specified demand is limited to one occurrence each twelve months or as otherwise agreed with Aurora Energy. Aurora requires two months notice for a specified demand reset.

### 7.0 Definitions

|                                                 |                                                                                                                                                                                              |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>afternoon boost</i></b>                   | An arranged time when a heating appliance may be connected to a controlled heating tariff at a time other than that specified as normal off peak.                                            |
| <b><i>Connection Point</i></b>                  | The agreed point of supply established between Aurora and the customer.                                                                                                                      |
| <b><i>Distribution Network</i></b>              | A network which is not a transmission network.                                                                                                                                               |
| <b><i>Distribution System</i></b>               | A distribution network, together with the connection assets associated with the distribution network, which is connected to a transmission system.                                           |
| <b><i>Distribution Use of System (DUoS)</i></b> | A charge to a distribution network user for use of the distribution system for the conveyance of electricity.                                                                                |
| <b><i>ICC</i></b>                               | Has the meaning “Individually Calculated Customer”.                                                                                                                                          |
| <b><i>Embedded Generation</i></b>               | Is a generation unit connected within a distribution system and not having direct access to a transmission system.                                                                           |
| <b><i>Irrigation</i></b>                        | The provision of water pumping capability that facilitates primary production.                                                                                                               |
| <b><i>Load Factor</i></b>                       | This is the ratio of a customer’s average electrical load divided by the maximum electrical load.                                                                                            |
| <b><i>Maximum Demand</i></b>                    | The electrical demand measured in kilo Watts (kW) and is calculated as the energy consumption recorded over the demand integration period divided by the demand integration period in hours. |

**National Electricity Market (or NEM)** As defined in the Tasmanian Electricity Code.

**Obsolete Tariffs** Tariffs that have been superseded but remain in place until such time as they are rescinded or the electrical configuration of a customer's installation is altered.

**Published Tariffs** Those network tariffs published from time to time, usually annually, by Aurora Energy.

**Residential Premise** A reference to a private residence, house, unit or apartment is a reference to an establishment that, in reasonable opinion of Aurora Energy is not classifiable under the Australian and New Zealand Standard Industrial Classification (ANZSIC). The ANZSIC system is based on businesses, and it is used to classify businesses. It includes any organisation which provides goods and services, including companies, non-profit organisations, government departments and enterprises.

**Retailer of Choice** The customer's current or chosen electricity retailer.

**Supply Voltage** The nominal voltage measured at the connection point.

**Time of Use** A tariff application that has variable rates dependent on the time of day electricity is consumed.

**Transmission Use of System** The charges associated with the services provided by a transmission system that are associated with the conveyance of electricity through the transmission system.

## 8.0 References

Aurora Energy Published Network Tariff Rates

[http://www.auroraenergy.com.au/powerline\\_network/quicklinks/network\\_tariffs.asp](http://www.auroraenergy.com.au/powerline_network/quicklinks/network_tariffs.asp)

Report on Formation and Introduction of Network Tariffs

[http://www.auroraenergy.com.au/powerline\\_network/quicklinks/network\\_tariffs.asp](http://www.auroraenergy.com.au/powerline_network/quicklinks/network_tariffs.asp)

Australian Energy Regulator

<http://www.aer.gov.au/>

National Electricity Market Management Company

<http://www.nemmco.com.au/>

Transend Networks Pty Ltd

<http://www.transend.com.au/>